Hossein Kavianirad

Doctoral Researcher in Control and Rehabilitation Robotics
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Research Profile

Control engineer working on **human-centred shared control** of wearable assistive systems. My doctoral research develops cooperative control for **hybrid FES-exoskeletons**, which combine functional electrical stimulation of the user's own muscles with robotic actuation, including EMG-based estimation of volitional effort under stimulation, participant-specific neuromuscular modelling, and dynamic allocation of assistance between muscles and motors. These methods were developed within the EU Horizon project *ReHyb* and evaluated with stroke patients in clinical settings. Earlier work covers boundary control of infinite-dimensional (PDE–ODE) systems, contact mechanics, and embedded control with sensor fusion.

Interests — Shared and cooperative control · hybrid FES-exoskeletons · rehabilitation and assistive robotics · EMG and biosignal processing · neuromuscular modelling · system identification · nonlinear and boundary control

Education

Ph.D. Candidate in Control — 2021 – present
Technical University of Munich (TUM), Chair of Information-oriented Control — *Munich, Germany*

- Dissertation: *Cooperative Shared Control of Hybrid FES-Exoskeleton*
- Advisor: Prof. Sandra Hirche
- Topics: shared control, system identification, exoskeletons, FES, EMG

M.Sc. in Mechanical Engineering (Dynamics and Control) — 2015 – 2017
Sharif University of Technology — *Tehran, Iran*

- GPA 19.27/20 (4.0/4.0), ranked 2nd of 151
- Thesis: *Boundary control of a flexible satellite (control design for hybrid PDE–ODE systems)*

B.Sc. in Mechanical Engineering — 2011 – 2015
Shiraz University — *Shiraz, Iran*

- GPA 19.05/20 (3.96/4.0), ranked 1st of 89
- Thesis: *Energy harvesting in a piezoelastic beam excited by air flow*

Research Experience

Research Assistant (Doctoral Researcher) — 2021 – present
Chair of Information-oriented Control, Technical University of Munich — *Munich, Germany*

- Research within the EU Horizon project *ReHyb* on hybrid FES-exoskeleton rehabilitation of the upper limb.
- Developed cooperative and model-based shared controllers that distribute assistance between FES and robotic actuation.
- Designed EMG-based estimation of volitional torque during stimulation (IEEE IECBES 2022 Best Paper Award).
- Designed and experimentally evaluated control strategies for human-centred systems with healthy participants and in clinical settings with stroke patients.

- Supervised more than 10 student projects (Bachelor's and Master's level).

Research Assistant – Control Engineer 2017 – 2021
Shiraz University Research Center *Shiraz, Iran*

- Control design and system identification for electromechanical systems.
- Actuator-in-the-loop, processor/sensor-in-the-loop and human-in-the-loop testing.
- Embedded implementation with microcontrollers, sensors and actuators; inertial navigation (INS), data fusion and Kalman filtering.

Publications

12 publications: 6 journal articles, 4 conference papers, 1 workshop paper, 1 preprint. Full and up-to-date list on Google Scholar. Earlier papers appear under the name *H. Kaviani Rad*.

Preprints

[P1] **H. Kavianirad**, S. Endo, D. Astarita, L. Amato, E. Trigili, S. Hirche. "Cooperative Control of Hybrid FES-Exoskeleton: Dynamic Allocation." *arXiv preprint* arXiv:2511.10117, 2025. [arXiv]

Journal Articles

[J1] L. Cazenave, C. Höhler, S. Endo, **H. Kavianirad**, et al. "Hybrid robot & FES rehabilitation after stroke: Short-term performance, fatigue, and patient experience." *Journal of Rehabilitation and Assistive Technologies Engineering*, vol. 13, 2026. [DOI]

[J2] C. Höhler, S. Endo, J. Hermsdörfer, L. Cazenave, **H. Kavianirad**, K. Jahn, C. Krewer. "Repetitive Grasping After Stroke Assisted by Functional Electrical Stimulation." *Artificial Organs*, 2025. [DOI]

[J3] **H. Kaviani Rad**, H. Salarieh, A. Alasty, R. Vatankhah. "Boundary control of flexible satellite vibration in planar motion." *Journal of Sound and Vibration*, vol. 432, pp. 549–568, 2018.

[J4] **H. Kaviani Rad**, H. Salarieh, A. Alasty, R. Vatankhah. "Boundary control of anti-symmetric vibration of satellite with flexible appendages in planar motion with exponential stability." *Acta Astronautica*, vol. 147, pp. 219–230, 2018.

[J5] **H. Kaviani Rad**, H. Nejat Pishkenari. "Frictional viscoelastic based model for spherical particles collision." *Granular Matter*, vol. 20, 2018. [DOI]

[J6] H. Nejat Pishkenari, **H. Kaviani Rad**, H. Jafari Shad. "Transformation of sliding motion to rolling during spheres collision." *Granular Matter*, vol. 19, 2017. [DOI]

Peer-Reviewed Conference Papers

[C1] **H. Kavianirad**, F. Missiroli, S. Endo, L. Masia, S. Hirche. "Toward Dexterous Hand Functional Movement: Wearable Hybrid Soft Exoglove-FES Study." *10th IEEE RAS/EMBS Int. Conf. on Biomedical Robotics and Biomechatronics (BioRob)*, pp. 1346–1351, 2024. [DOI]

[C2] N. Das, S. Endo, **H. Kavianirad**, S. Hirche. "Framework for Learning a Hand Intent Recognition Model from sEMG for FES-Based Control." *10th IEEE RAS/EMBS Int. Conf. on Biomedical Robotics and Biomechatronics (BioRob)*, pp. 1320–1327, 2024. [DOI]

[C3] **H. Kavianirad**, M. Forouhar, H. Sadeghian, S. Endo, S. Haddadin, S. Hirche. "Model-Based Shared Control of a Hybrid FES-Exoskeleton: An Application in Participant-Specific Robotic Rehabilitation." *IEEE Int. Conf. on Rehabilitation Robotics (ICORR)*, pp. 1–6, 2023. [DOI]

[C4] **H. Kavianirad**, S. Endo, T. Keller, S. Hirche. "EMG-Based Volitional Torque Estimation in Functional Electrical Stimulation Control." *IEEE-EMBS Conf. on Biomedical Engineering and Sciences (IECBES)*, pp. 171–176, 2022. **Best Paper Award.** [DOI]

Workshop Papers

[W1] A. Sena, **H. Kavianirad**, S. Endo, E. Burdet, S. Hirche. "The Gap in Functional Electrical Stimulation Simulation." *3rd Workshop on Closing the Reality Gap in Sim2Real Transfer for Robotics, Robotics: Science and Systems (RSS)*, 2022.

Talks

Cooperative Shared Control of Hybrid FES-Exoskeleton Jul 2026
Doctoral seminar talk, Chair of Information-oriented Control, TUM, Munich, Germany.

Honours and Awards

Best Paper Award, IEEE-EMBS IECBES 2022 2022
For *"EMG-Based Volitional Torque Estimation in Functional Electrical Stimulation Control"*, 7th IEEE-EMBS Conference on Biomedical Engineering and Sciences.

Teaching Assistantship funded by Iran's National Elites Foundation 2015 – 2017

Iran's National Elites Foundation Fellowship 2014 – 2015

Distinguished Student, Mechanical Engineering, Sharif University of Technology 2015 – 2016

Distinguished Student, Mechanical Engineering, Shiraz University 2014 – 2015

Top 0.3% of over 350,000 candidates, Iranian National University Entrance Exam 2011

Teaching and Supervision

Supervision	Supervised more than 10 Bachelor's and Master's student projects and theses at TUM (2021 – present) in control, dynamic modelling, learning and rehabilitation robotics.
Teaching	Teaching Assistant, Sharif University of Technology (2015 – 2017), funded by Iran's National Elites Foundation.

Research Projects

ReHyb	EU Horizon project on a hybrid FES-exoskeleton system for upper-limb rehabilitation after stroke, featured in a TUM press release (2025). Role: shared-control design, EMG-based estimation, experimental and clinical evaluation.

Technical Skills

Control	Shared and cooperative control, model-based and nonlinear control, boundary control of PDE systems, Kalman filtering and data fusion, system identification
Experimental	Exoskeletons and soft exogloves, functional electrical stimulation, surface EMG, human-in-the-loop and hardware-in-the-loop experiments, clinical studies
Programming	MATLAB, C/C++, microcontroller programming

References

Available upon request.